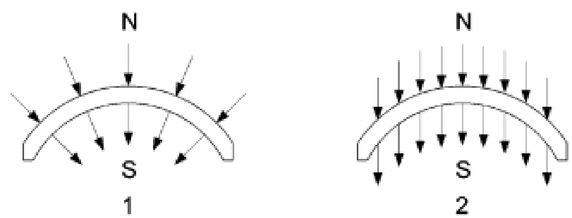
NUOVO DUCATO X250 2.2 JTD STARTER MOTOR AND COMPONENTS 5520B

SPECIFICATIONS

The starting system is composed of the ignition switch controlled by the key and the starter motor which is connected directly to the battery.

CIRCULAR MAGNET WITH MAGNETICALLY PREORIENTATED CRYSTALS

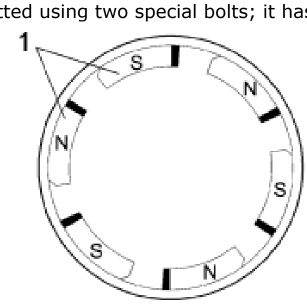
The rough components are directed into the desired shape by pressing and extrusion. Lastly, after a thorough baking procedure, the material turns into the magnets that have a black ceramic structure. The magnetizing treatment takes place during the manufacturing process.
The magnets have a circular sector and thereby replace the polar part and the winding.



1 - Radial orientation
2 - Parallel orientation

STATOR-INDUCTOR

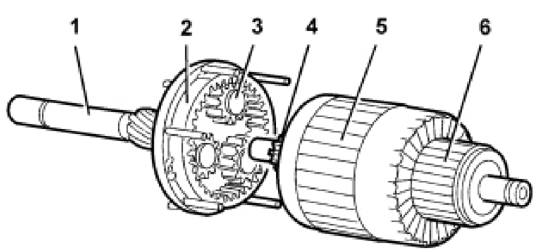
The positioning of the magnets cannot be done through strong compression or by means of riveting because the magnets are reasonably fragile: they are secured by flexible means (clips) which, in some cases, are stuck to the casing. The casing assembly can be compared with a simple mechanical component without wires to be fitted using two special bolts; it has 6 magnetic poles.



1 - Permanent N-S magnets

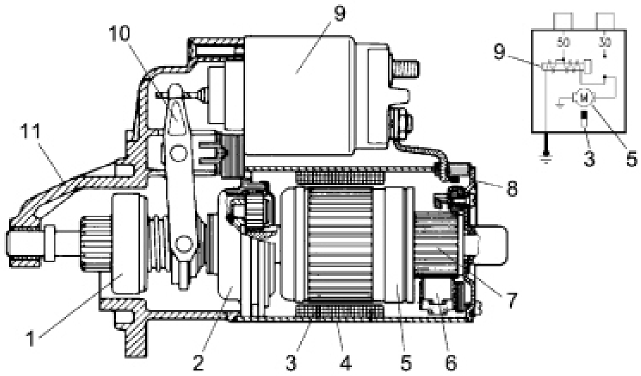
ROTOR-ARMATURE (WITH EPICYCLIC REDUCTION UNIT)

The rotor-armature is illustrated in the diagram below.



1 - Shaft;
2 - Ring gear/crown wheel;
3 - Satellite/planet gears;
4 - Pinion/sun gear;
5 - Armature;

6 - Commutator/switch;
Given that the magnetic field is produced by permanent magnets, the energizing is the same in all motor operating conditions; in addition, as there are no energizing windings, there is no energizing current or ohmic resistance: the total resistance only depends on the armature circuit. The rotation speed is only influenced by the current that passes through the blades and reaches the rotor depending on the voltage.
Since the motor is equipped with permanent magnets, i.e. with fixed polarity, the direction of rotation depends on how the current is sent to the blades.
When rotating the rotor transmits motion to the three satellite gears fitted on special supports that transmit their drive torque to the ring gear or crown wheel in one piece with the output shaft where the pinion is located and the engine flywheel is rotated. The optimum reduction radio 1:4.26 is 4.26 revs of the rotor for every revolution of the ring gear and the presence of permanent magnets makes it possible to produce small simply designed motors with a comparable power output in relation to motors with windings.



1 - Pinion with free wheel
2 - Epicyclic reduction unit
3 - Inductor permanent magnets
4 - Polar casing assembly
5 - Armature or rotor
6 - Blades
7 - Commutator
8 - Commutator side support
9 - Engagement relay
10 - Lever
11 - Control side support

TECHNICAL DATA

The types of starter motor adopted on this vehicle are summarized in the table below.

Starter motor	2.2 JTD	2.3 JTD	3.0 JTD
Manufacturer	BOSCH	FPT	BOSCH
Make and type	R78 - M28	DRX2.3	R78 - M55
Voltage [V]	12	12	12
Nominal power [KW]	-	2.3	-
Rotation	Clockwise	Clockwise	Clockwise